

MB 311: Management Fundamentals for Engineers

Course Code	MB 311
Course Name	Management Fundamentals for Engineers
L-T-P-C	3-0-0-3
Prerequisites	Nil
Intended for	Minor in Management
Distribution	Core Level: 300
Approval	64th BoA (May 2026), in the Minor in Management curriculum

Preamble

This course is designed specifically for engineering students and is distinct from conventional MBA foundation courses. It aims to develop managerial literacy relevant to technology-driven and innovation-oriented environments by introducing students to the foundations of management, organizations, leadership and strategy in contemporary engineering and business contexts. The course emphasizes interdisciplinary collaboration, product thinking, innovation ecosystems, and organizational decision-making under uncertainty. It focuses on how organizations function, how technical teams are managed, how technologies and products are commercialized, and how engineers engage with complex organizational and market systems. The pedagogy integrates technology-oriented examples, simulations, analytical problem-solving, and real-world applications relevant to engineers and technology.

Course Objectives

- Introduce students to managerial thinking in technology-driven organizations.
- Develop understanding of organizational systems, leadership, and team dynamics in engineering contexts.
- Strengthen communication, collaboration, and problem-solving skills for professional settings.

Course Contents

- **Management in Technology Organizations:** Evolution and functions of management; Technology-driven organizations and innovation ecosystems; Roles of managers in engineering and product environments; Systems thinking and interdisciplinary collaboration; Management in VUCA world and technology changes. (10 hours)
- **Organizational Behaviour, Teams, and Leadership:** Organizational structures and culture; Managing technical and cross-functional teams; Motivation and team dynamics; Leadership, communication, and conflict management in technology organizations. (8 hours)
- **Managerial Decision-Making and Problem Solving:** Decision-making under uncertainty; Analytical and data-driven thinking; Problem-solving frameworks; Managerial trade-offs and stakeholder perspectives. (8 hours)
- **Strategy, Innovation, and Technology Commercialization:** Competitive advantage and business models; Innovation management and product thinking; Technology commercialization and platform ecosystems; Startup ecosystems and entrepreneurial environments. (8 hours)
- **Digital Transformation and Emerging Organizational Challenges:** AI-enabled organizations and digital transformation; Ethics and sustainability in technology environments; Future of work and changing organizational systems; Emerging challenges in technology and society. (8 hours)

Laboratory Modules

None

Text Books

1. Robbins, S. P., & Coulter, M. *Management*. Pearson.

References

1. Drucker, P. F., *Management: Tasks, Responsibilities, Practices*. Harper Business.